

ENCARDIO-RITE

METASENSING FOR MEGA STRUCTURES



GNSS Receiver Base & Rover Configuration USER MANUAL

Web-based Configuration Interface

Doc. # WI 6002.xxx Rev. 00
September 2026

Encardio-Rite Electronics Pvt. Ltd. | Lucknow, India | www.encardio.com

Contents

Contents.....	2
1 Introduction	3
1.1 About this manual.....	3
1.2 System overview	3
1.3 Conventions used in this manual	3
2 Front Panel Overview.....	3
2.1 Controls.....	4
2.2 Status indicators (top row).....	4
2.3 Power & system indicators (bottom row).....	4
3 Accessing the Configuration Interface	5
3.1 Physical setup.....	5
3.2 Set the PC network address	5
3.3 Log in to the web interface	5
4 Configuring the Receiver as a BASE	7
4.1 Base coordinates – Auto Fix.....	7
4.2 Base coordinates – Manual.....	9
4.3 Operating mode	11
4.4 Server (NTRIP caster) settings.....	11
4.5 FTP settings	13
4.6 Save and finish	13
5 Configuring the Receiver as a ROVER.....	15
5.1 Receiver type, operating mode and server settings	15
5.2 FTP and rover settings.....	17
5.3 Scan time, deviation axis, storage and reporting.....	19
5.4 Save	20
6 Verifying Operation.....	20
7 Troubleshooting	20
8 Safety & Support	20

1 Introduction

1.1 About this manual

This manual explains how to configure the Encardio-Rite GNSS receiver as a Base station or as a Rover using its built-in web configuration interface. It covers connecting to the receiver, logging in, and setting every parameter required for reliable RTK operation and data reporting.

The receiver is configured entirely through a web browser — no separate PC software needs to be installed.

1.2 System overview

A typical deployment uses at least one Base receiver and one or more Rover receivers:

- The BASE is installed over a known, stable point. It computes (or is given) its own precise position and streams RTK correction data (RTCM) to an NTRIP caster.
- Each ROVER connects to the caster, applies the corrections, and computes its own position to centimetre-level accuracy. It then reports position/deviation data and uploads log files to an FTP server.

Both receiver types are set up from the same web interface; the on-screen options change according to the selected Receiver Type.

1.3 Conventions used in this manual

- Bold text refers to an on-screen label, button, or field name — for example, Save Configuration.
- Values shown in the screenshots (IP addresses, ports, IDs) are examples only. Replace them with the values for your site.

NOTE The receiver clock and all logging times use UTC. Enter Scan Start Time in UTC, not local time.

2 Front Panel Overview

The front panel carries the power button, two rows of status LEDs, and a Bluetooth/Wi-Fi indicator.



Fig. 2.1 GNSS receiver front panel

2.1 Controls

Control	Function
PWR	Power on / off button. Press to switch the receiver on; press and hold to switch off.
BT / WIFI	Bluetooth / Wi-Fi indicator and activation. Indicates when the wireless interface is active for local access.

2.2 Status indicators (top row)

LED	Indication
PPS	Pulse-Per-Second from the GNSS engine. A steady blink indicates the receiver has valid satellite timing.
SAT	Satellite tracking. Lit / blinking when the receiver is tracking a sufficient number of satellites.
RTK	RTK solution status. Indicates when a fixed RTK (centimetre-level) solution is achieved.
NET	Network / NTRIP link. Lit when the receiver is connected to the network / caster.
COM	Communication activity between the GNSS module and the on-board computer.

2.3 Power & system indicators (bottom row)

LED	Indication
DC IN FUSE	Health of the DC-input fuse. Lit = external DC supply present and fuse healthy.
BATT FUSE	Health of the battery fuse.
BKUP MODE	Backup mode. Lit when the unit is running from the internal / backup battery.
BATT LOW	Battery low warning. Lit when the battery voltage is below the safe threshold.

LED	Indication
SYS STAT	Overall system status / heartbeat.

NOTE LED colours and blink patterns are indicative. Use the web interface (Section 6) to confirm exact status.

3 Accessing the Configuration Interface

3.1 Physical setup

1. Mount the GNSS antenna on a stable structure with a clear, obstruction-free view of the sky, and connect the antenna cable to the receiver.
2. Connect power: use a 12 V DC adapter on the DC-input connector, or a battery source.
3. Connect the receiver to the configuring PC with an Ethernet cable and switch the receiver on with the PWR button.

3.2 Set the PC network address

The PC must be on the same subnet as the receiver. If the receiver address is 192.168.10.100, set the PC to a free address on that subnet, for example 192.168.10.50.

4. Open Control Panel → Network and Sharing Center → Change adapter settings.
5. Right-click the Ethernet connection → Properties → Internet Protocol Version 4 (TCP/IPv4) → Properties.
6. Select **Use the following IP address** and enter, for example, IP 192.168.10.50 and Subnet mask 255.255.255.0. Click OK.

3.3 Log in to the web interface

7. Open a web browser and go to the receiver address, for example <http://192.168.10.100>
8. Enter the Username and Password and click Login (default credentials are provided with the device).

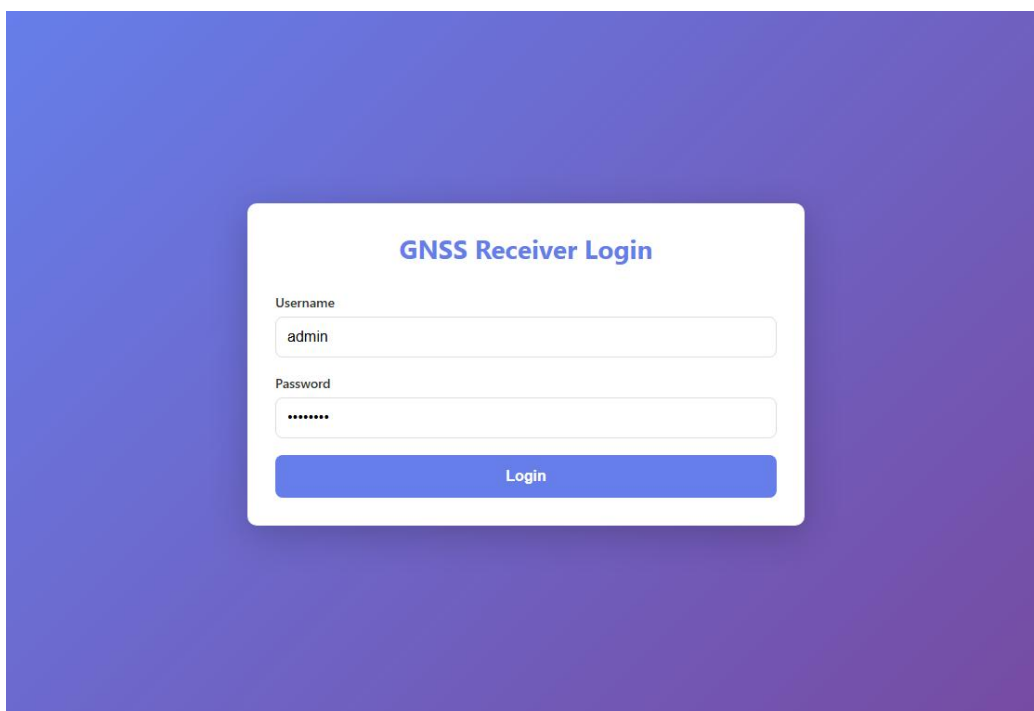


Fig. 3.1 Login screen

After logging in, the GNSS Receiver Setup page opens. The first setting, Receiver Type, determines whether the unit is configured as a Base or a Rover.

4 Configuring the Receiver as a BASE

Select BASE under Receiver Type. The Base Co-ordinates options appear. A base can either determine its own position automatically (Auto Fix) or use coordinates you enter manually.

4.1 Base coordinates – Auto Fix

In Auto Fix mode the base averages its own position over a chosen period and then locks that value as its reference coordinate.

GNSS Receiver Setup

User

Server LogsNormal LogsLogout

GNSS Receiver Settings

Receiver Type

☒ BASE☐ ROVER☐ STOP

Base Co-ordinates

☒ Auto Fix Base Co-ordinates☐ Manually feed Base Co-ordinates

☐ Reset Base Co-ordinates

Average Delay (Change requires base reset.)

1 hour

GNSS Operating Mode

☐ Continuous Mode☐ Burst Mode

Diagnostics

☐ Enable Debug Logging (verbose logs to system journal)

Server Settings

Fig. 4.1 BASE – Auto Fix coordinates and operating mode

9. Select BASE, then choose Auto Fix Base Co-ordinates.

10. Set **Average Delay** – the averaging duration (1–72 hours). A longer period gives a more accurate fixed position.
11. Tick **Reset Base Co-ordinates** only when you want the base to discard its stored position and re-average a fresh one.

NOTE Changing the Average Delay takes effect only after a base reset (tick Reset Base Co-ordinates). Once the base has locked a position, changing the delay alone has no effect until it is reset.

4.2 Base coordinates – Manual

If the exact coordinates of the monument are already known (from a survey), enter them directly.

GNSS Receiver Setup

User

Server LogsNormal LogsLogout

GNSS Receiver Settings

Receiver Type

☒ BASE☐ ROVER☐ STOP

Base Co-ordinates

☐ Auto Fix Base Co-ordinates☒ Manually feed Base Co-ordinates

Latitude

26.84670000

Longitude

80.94620000

Altitude

123.456

GNSS Operating Mode

☐ Continuous Mode☐ Burst Mode

Fig. 4.2 BASE – Manually fed coordinates

12. Select Manually feed Base Co-ordinates.
13. Enter Latitude, Longitude and Altitude for the base monument.

4.3 Operating mode

Choose how the receiver runs:

- Continuous Mode – the receiver stays powered and streams / logs data continuously.
- Burst Mode – the receiver wakes on a schedule, operates for a set window, then sleeps to save power. Set the burst window and interval when this mode is selected.

4.4 Server (NTRIP caster) settings

These settings tell the base where to publish its RTCM correction stream.

IP Server

192.168.10.50

Port Server

2101

Mount Point

BASE01

User

base1

Password

.....

FTP Settings

☐

FTP Upload Enable

FTP IP

192.168.1.100

FTP Port

21

FTP User

FTP Password

Fig. 4.3 BASE – Server and FTP settings

Field	Description	Example
IP Server	IP address of the NTRIP caster the base streams to.	192.168.10.50
Port Server	Caster port.	2101
Mount Point	Mount point name the base publishes on.	BASE01
User	Caster user name for the base.	base1
Password	Caster password.	•••••

4.5 FTP settings

When enabled, the base uploads its configuration backup to the FTP server so it can be restored to replacement hardware.

Field	Description
FTP Upload Enable	Turn FTP upload on / off.
FTP IP	FTP server address.
FTP Port	FTP server port (typically 21).
FTP User / Password	FTP account credentials.

4.6 Save and finish

The screenshot displays a configuration interface with a white background and purple vertical bars on the left and right sides. At the top, there is a blue button labeled "Save Configuration". Below this, the "WiFi Settings" section is highlighted with a blue header and a horizontal line. It contains two input fields: "WiFi SSID" with the placeholder text "Enter WiFi network name" and "WiFi Password" with the placeholder text "Enter WiFi password". Below these fields is another blue button labeled "Save WiFi Settings". The "IP Settings" section follows, also with a blue header and a horizontal line. It includes two input fields: "IP Address" with the placeholder text "e.g., 192.168.1.60" and "Port Number" with the value "2000".

Fig. 4.4 Save, Wi-Fi and IP settings

14. Click **Save Configuration** to apply the base settings.
15. Optionally configure Wi-Fi Settings (SSID / password) for wireless access and click Save WiFi Settings.
16. Optionally change the receiver IP Settings and click Save IP Settings. The receiver will then use the new address.

5 Configuring the Receiver as a ROVER

Select ROVER under Receiver Type. The Rover Settings, Data Storage and Reporting options appear.

5.1 Receiver type, operating mode and server settings

GNSS Receiver Setup

SN: Not Updated

User

Server Logs

Normal Logs

Logout

GNSS Receiver Settings

Receiver Type

☐ BASE

☒ ROVER

☐ STOP

GNSS Operating Mode

☐ Continuous Mode

☐ Burst Mode

Diagnostics

☐ Enable Debug Logging (verbose logs to system journal)

Server Settings

IP Server

192.168.10.50

Port Server

2101

Mount Point

BASE01

Fig. 5.1 ROVER – receiver type, operating mode and server settings

17. Select ROVER.

18. Choose the Operating Mode (Continuous or Burst) as described in Section 4.3.

19. Under Server Settings, enter the NTRIP caster the rover receives corrections from:

Field	Description	Example
IP Server	NTRIP caster IP address.	192.168.10.50
Port Server	Caster port.	2101
Mount Point	Mount point of the base to receive corrections from.	BASE01
User / Password	Caster credentials for the rover.	rover1 / ●●●

5.2 FTP and rover settings

The screenshot displays a configuration interface for a rover system, organized into several sections. The 'User' section contains a text field with 'rover1'. The 'Password' section contains a masked field with eight dots. The 'FTP Settings' section is highlighted with a blue header and includes a checked checkbox for 'FTP Upload Enable', followed by fields for 'FTP IP' (203.0.113.10), 'FTP Port' (21), 'FTP User' (ftpuser), and 'FTP Password' (masked with six dots). The 'Rover Settings' section, also highlighted with a blue header, includes fields for 'Rover ID' (ROVER_01) and a 'Scan Interval' dropdown menu currently set to '1 Minute'.

User

rover1

Password

.....

FTP Settings

☒ FTP Upload Enable

FTP IP

203.0.113.10

FTP Port

21

FTP User

ftpuser

FTP Password

.....

Rover Settings

Rover ID

ROVER_01

Scan Interval

1 Minute ▼

Fig. 5.2 ROVER – FTP and rover settings

Under FTP Settings, enable upload and enter the FTP server details so the rover can upload its logged data files.

Field	Description	Example
Rover ID	Unique identifier for this rover; used in data file names.	ROVER_01
Scan Interval	How often a data record is logged / reported.	1 Minute

5.3 Scan time, deviation axis, storage and reporting

Scan Start Time (UTC – System uses UTC time)

09:00:36

HH

MM

06 ▾

00 ▾

Deviation Reporting Axis

☐ Report deviation along & transverse to a reference axis i

Data Storage

Max Record Per File

1 records ▾

Upload File Action

Delete After Upload ▾

Date Time Format

yyyy/MM/dd HH:mm:ss ▾

Reporting

☒ GeoDetic Data Upload Enable

Fig. 5.3 ROVER – scan start, deviation axis, storage and reporting

Field	Description
Scan Start Time (UTC)	Time of day (HH:MM, UTC) from which logging is scheduled.
Deviation Reporting Axis	When enabled, deviation is reported along and transverse to a reference axis; enter the axis azimuth.
Max Record Per File	Number of records stored in each data file before a new file is started.

Field	Description
Upload File Action	What to do with a file after successful FTP upload – Delete or Archive.
Date Time Format	Timestamp format used in data files (e.g. yyyy/MM/dd HH:mm:ss).
GeoDetic Data Upload Enable	Include geodetic (lat/lon/height) values in the uploaded data.
Deviation Unit	Unit for reported deviation values (e.g. mm).

5.4 Save

20. Click **Save Configuration** to apply the rover settings. Configure Wi-Fi and IP settings if required, as in Section 4.6.

6 Verifying Operation

After saving, confirm the receiver is working correctly:

- On the front panel, check that SAT indicates satellite tracking and, for RTK operation, that RTK shows a fixed solution and NET shows a caster connection.
- In the web interface, open Server Logs or Normal Logs (top-right of the setup page) to review recent activity and confirm data is being logged and uploaded.

7 Troubleshooting

Symptom	Checks / action
Cannot open the web page	Confirm the PC IP is on the receiver's subnet (Section 3.2), the Ethernet cable is connected, and the receiver is powered on.
No RTK fix on the rover	Check Server Settings (caster IP/port/mount/credentials) and that the base is streaming; confirm satellite visibility.
No data on the FTP server	Confirm FTP Upload is enabled and the FTP IP/port/credentials are correct; check the logs for upload errors.
Base position keeps changing	Ensure Reset Base Co-ordinates is not left ticked; once a good position is locked, untick it.
Weak / intermittent signal	Check the antenna mounting, cable and sky visibility; relocate the antenna away from obstructions.

8 Safety & Support

- Use only the specified DC supply and a correctly rated battery.
- Install the antenna and mast safely; observe site electrical-safety rules and keep clear of overhead power lines.
- Protect the receiver from water ingress beyond its rated protection and from extreme temperatures.

For support, contact Encardio-Rite: geotech@encardio.com | www.encardio.com